

Knowledge, technology, science

Sept 30

Migration, Knowledge Diffusion and the Comparative Advantage of Nations

Dany Bahar and Hillel Rapoport, 2018

Research question:

How does migration shape a country's export structure and revealed comparative advantage (RCA)?

Data:

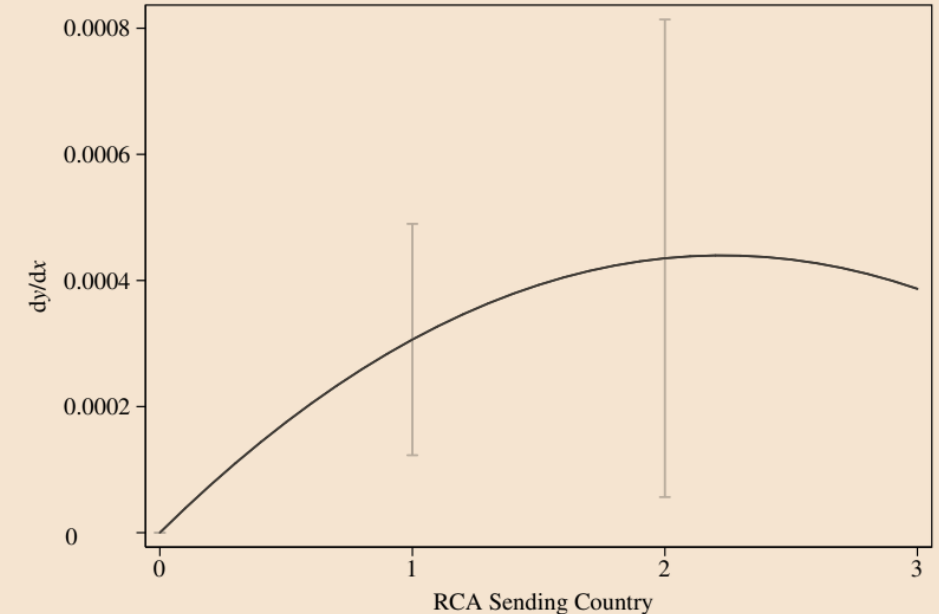
- 135 countries, 781 product groups (SITC-4)
- Timeframe: 1990–2010 (two 10-year periods)
- Sources: UN Comtrade, Artuc et al. (2015), OECD FDI

Methodology:

The impact of the migrant population weighted by the RCA of countries of origin...

- ...on the emergence of new export products
⇒ Probit model
- ...on the growth of existing exports
⇒ OLS panel model

Diminishing Marginal Returns to RCA of Partner Countries



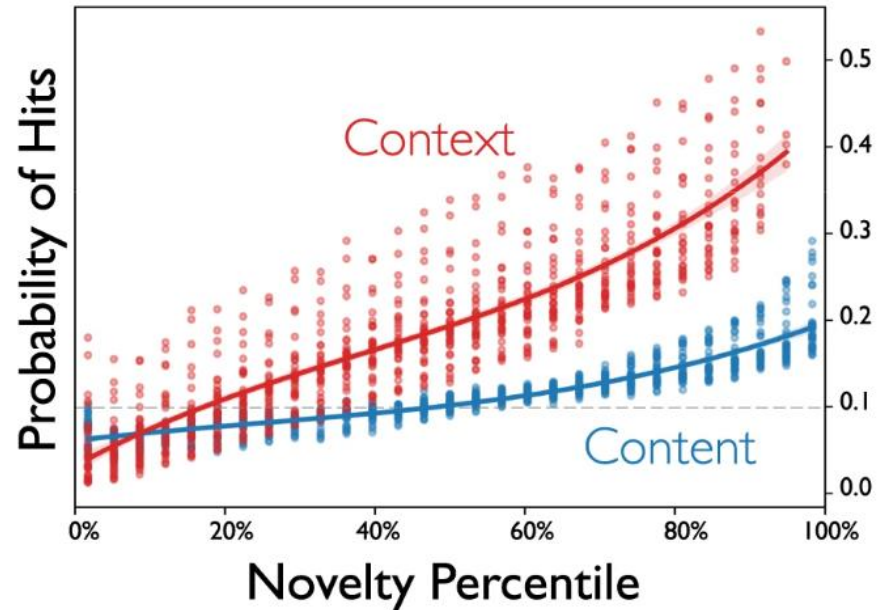
Results:

- **Main Effect:** +10% immigrants
⇒ +2% **probability** of exporting a new product
- **Skill Level:** High-skilled impact is **4-10x stronger** (> 100 000 USD FDI)
- **Mechanism (Graph):**
 - Zero effect if origin RCA = 0
⇒ proves **product-specific tacit knowledge**
 - Diminishing returns; peak impact at RCA ≈ 2

Surprising research and the impact of scientific outsiders

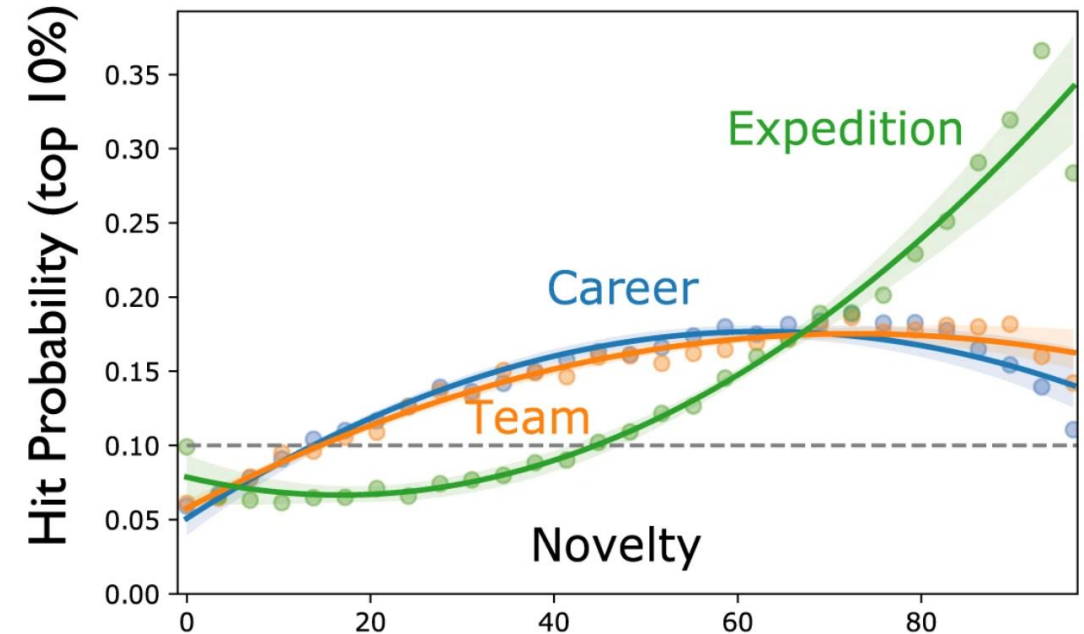
Which knowledge combinations attract high impact—and who produces them?

1. Is surprising research more influential?



- **Content:** unusual combinations of concepts and methods
- **Context:** unusual combinations of cited disciplinary literatures
- **Data:** 20+M articles & 6.5M patents
- **Method:** Hypergraph model learns expected combinations; unlikely combinations score as surprising

2. Where does influential research come from?



- **Career:** unusual disciplinary background
- **Team:** unusual mix of researchers' backgrounds
- **Expedition:** publishing in a field distant from researchers' previous work

High impact is especially associated with unexpected disciplinary combinations and researchers entering distant fields.

Technological Change and Political Polarization: Automation Risk and Radical Right Voting

Anelli, Colantone, & Stanig (2021), PNAS

Zorka Albert

Research Problem & Question

Problem: The change toward industrial robotics and automation reshapes labor markets unevenly. It displaces routine-task workers while rewards high-skill capital.

Question: Does exposure to labor-replacing automation raises an individual's tendency to vote for radical right parties?

Connection to Course: Shows that technological change and skill-biased economic changes produce structural polarization that leads to electoral instability.

Data & Identification Strategy

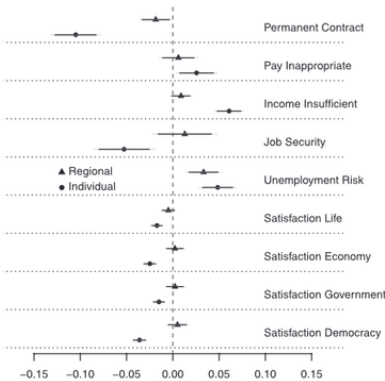
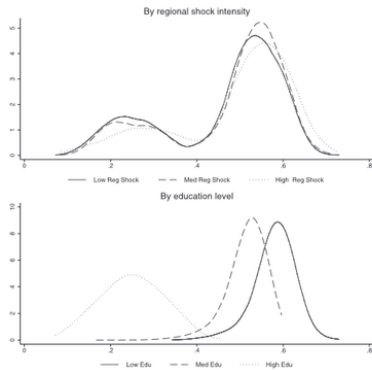
Data Sources:

- **Individual Data:** Multi-wave ESS tracking voting choices across 13 Western European countries (1999-2015).
- **Automation Data:** Industry operational stock of industrial robots (IFR).

Methodology:

- Constructed an Individual Robot Vulnerability Index (RVI): individual's occupation, education, routine task intensity, and industry-level robot penetration.
- Controlled for globalization exposure (baseline socio-economic characteristics, regional fixed effects, and general import competition).

Key Visuals



Workers in routine occupations threatened by robotization show a statistically significant rise in support for radical right parties, driven by economic insecurity and cultural backlash.

Key Findings & Stability Implications

Direct Electoral Impact: Increased vulnerability to robot adoption increases the probability of voting for radical right parties, controlling for traditional trade exposure.

Dual Mechanisms: Dissatisfaction is expressed both via economic insecurity (threat of displacement, stagnant wages) and cultural defensiveness (status insecurity, anti-immigrant attitude).

Systemic Risk & Institutional Stability: Technological shocks erode political unity; where improvements in technology occur faster than improvements in skill and geographic adjustment by workers, polarization is facilitated.

FOLLOW THE MONEY: A STARTUP-BASED MEASURE OF AI EXPOSURE ACROSS OCCUPATIONS, INDUSTRIES, AND REGIONS

Research problem and question

Technical capability alone does not explain commercial development. Which occupations do funded AI startups target?

Data

The study combines 1,016 O*NET job descriptions with 958 Y Combinator AI startups funded in 2005-March 2024. It checks robustness using 830 European startups.

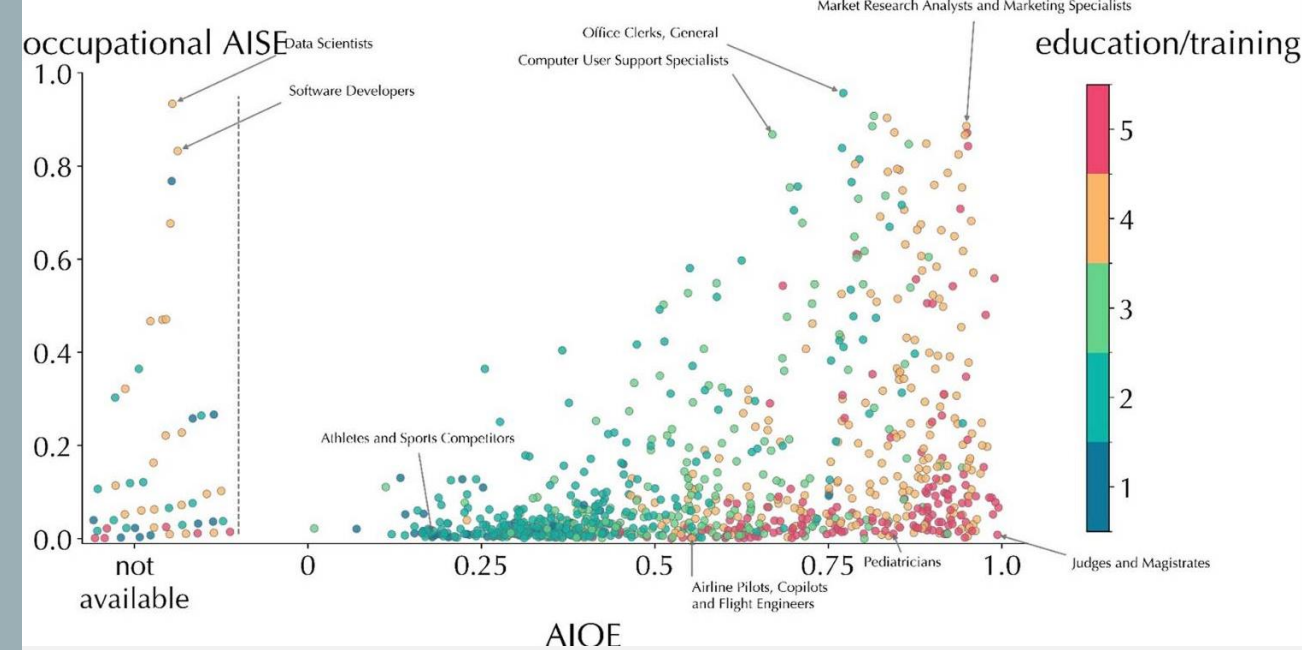
Method

Llama 3 8B (LLM) judges whether each startup's product could replace any described job tasks. AI Startup Exposure (AISE) is the share of startups matched to an occupation.

Main finding and interpretation

High technical exposure does not automatically translate into strong startup targeting. E.g. office clerks and data scientists attract significant startup activity, while judges and pediatric surgeons receive less attention despite their technical exposure. This suggests that AI's commercial development depends on market incentives and professional responsibility as well as technological capability. Assessments of potential workplace change should therefore consider both what AI can do and what businesses are developing.

Limit: AISE measures startup targeting, not actual adoption or job losses. It misses innovation outside the sampled startups.



AIOE (AI Occupational Exposure, horizontal axis) measures potential exposure based on the overlap between AI capabilities and the abilities a job requires. Further right means higher technical exposure.

AISE (AI Startup Exposure, vertical axis) measures the share of sampled AI startups whose products match an occupation's tasks. Higher values mean more startup targeting.

Figure:

Each dot represents an occupation. Colors indicate education and training requirements, from 1 (little preparation) to 5 (extensive preparation). The separate left column contains occupations without AIOE scores. Both indicators are rescaled to 0–1, rather than representing probabilities of job loss.

Observations:

Among occupations with high technical exposure, startup targeting varies widely. Office clerks appear near the top right, while judges appear near the bottom right. This gap suggests that commercial incentives, trust and responsibility also shape which occupations startups target.

Source: Fenoaltea et al. (2026). Follow the money: A startup-based measure of AI exposure across occupations, industries, and regions. PNAS Nexus, 5(6), pgag185.

<https://doi.org/10.1093/pnasnexus/pgag185>

Specialization

October 7

Why did some countries catch-up, while others got stuck in the middle? Stages of productive sophistication and smart industrial policies

Dominik Hartmann, Ligia Zagato, Paulo Gala, Flavio L

Research question

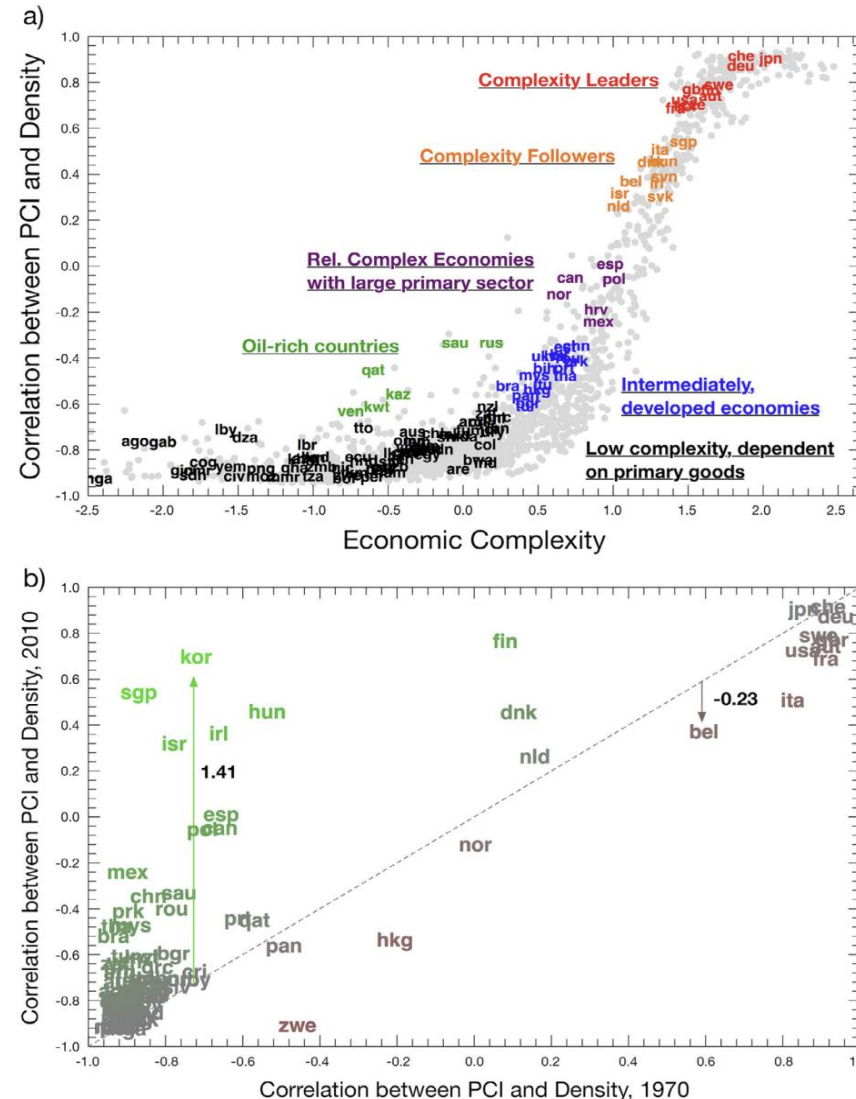
Why do some middle-income countries successfully transform toward complex production while others remain stuck in simple activities?

Data

- 116 countries, 1970–2010
- International trade data
- 774 products
- Economic & Product Complexity

Method

- Economic Complexity Index (ECI)
- Product Space & product proximity
- Closeness to simple vs. complex products
- Country case studies / industrial policies



Key findings

- Most countries fail to cross the intermediate stage.
- Only 5 countries fully transformed: Ireland, Israel, Hungary, Singapore, and South Korea.
- Successful countries exploited opportunities in digital & electronics.
- Smart industrial policies helped build domestic capabilities and access international knowledge.

Takeaway

Development requires structural transformation toward complex products — not simply higher output.

Relatedness and regional specialization in green, digital, and twin economic activities

Evidence from the UK

Cicerone, Losacker & Ortega-Argilés (2026), The Annals of Regional Science

1. Research Question

Can relatedness between green and digital activities shape regional specialization?
Can “twin” activities provide a bridge between the two domains?

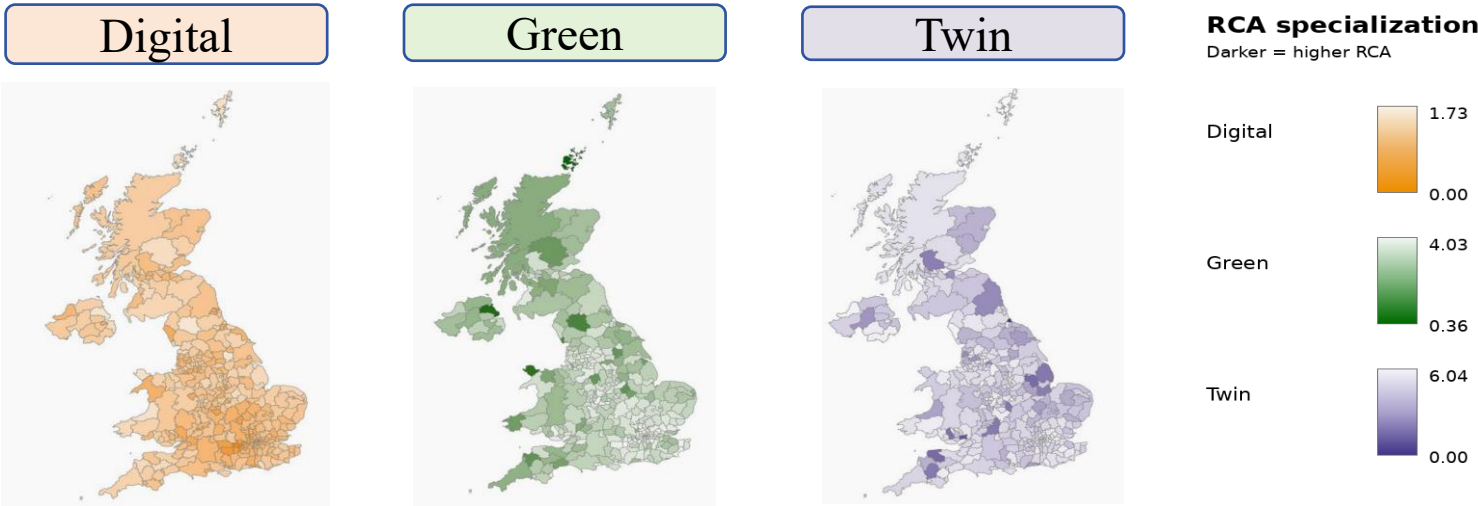
2. Research Data

- ~ 200,000 UK firms (web-based data)
- 385 emerging-industry subsectors
- 374 UK local authorities
- Classified as: **green** (45), **digital** (166), **twin** (13) subsectors using machine learning and expert classification

3. Research Method

- Regional specialization: RCA
- Industry relatedness: firm-level co-occurrence
- Relatedness density: proximity to existing activities
- OLS regressions with regional and industry controls

4. Regional specialization in digital, green, and twin industries



5. Key empirical findings

- Urban scaling : Digital $\beta=1.276$; Green $\beta=0.918$; Twin $\beta=0.887$
 - Cross-domain relatedness : How are the three domains connected?
- Twin activities may act as a bridge between green and digital capabilities.**

Cross-domain relatedness

Green relatedness → Digital RCA	0.053***	Digital relatedness → Green RCA	0.016***
Twin relatedness → Digital RCA	0.344***	Twin relatedness → Green RCA	0.233***

Table 3 coefficients; *** p < 0.01. Source: Cicerone, Losacker & Ortega-Argilés (2026).

6. Main Takeaway

Green and digital activities are positively related across UK regions, while twin activities show the strongest associations with both domains. This suggests that twin activities may provide a bridge between green and digital regional capabilities.

Economic complexity of prefectures in Japan

Chakraborty, Inoue & Fujiwara (2020)
<https://doi.org/10.1371/journal.pone.0238017>

What can industrial structure tell us about regional economic performance?

RESEARCH QUESTION

Can a non-monetary index of industrial complexity capture differences in prefectural economic performance?

KEY FINDING

Higher ECI is strongly associated with higher income and output.

DATA + METHOD

1,033,518 active firms, July 2016
 47 prefectures × 91 industrial sectors
 Hidalgo & Hausmann's Method of Reflections →
 Annual sales → $RCA \geq 1$ →
 prefecture-industry network → ECI.
 Fitness-complexity algorithm used as a robustness check.

MAIN RESULT

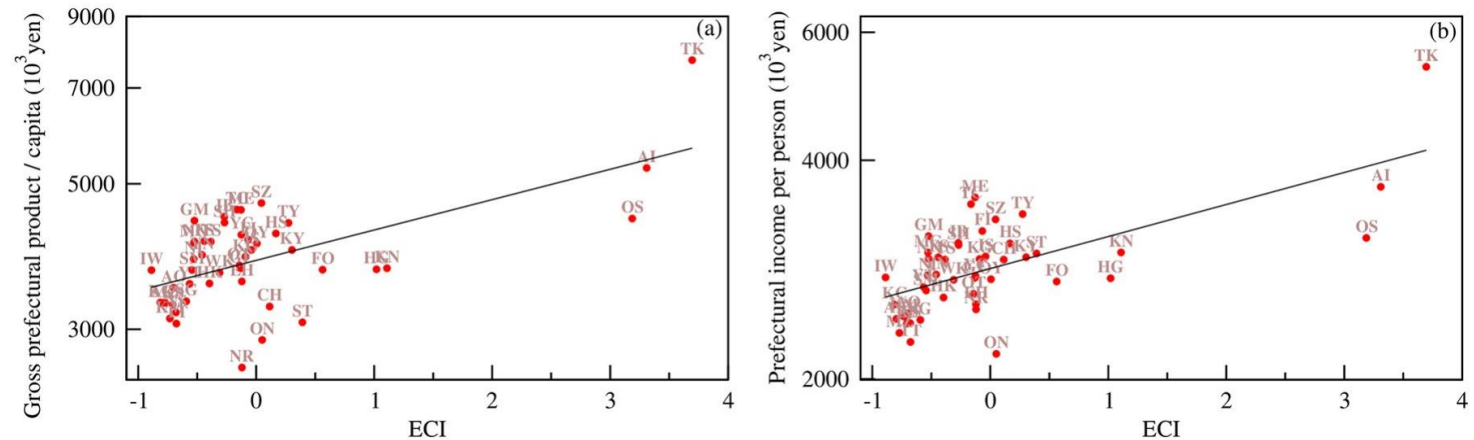
Prefecture-level correlations

$r = 0.661$ ECI ↔ GPP per capita
 $p = 4.2 \times 10^{-7}$

$r = 0.668$ ECI ↔ income per person
 $p = 9.0 \times 10^{-8}$

Fitness gives similar correlations: $r = 0.742$ and 0.746 .

PREFECTURE-LEVEL EVIDENCE



Why did chip manufacturing concentrate in East Asia?

Demand pulled capacity there, not just state sponsorship

The research problem

- East Asia's share of the world chip-fab capacity: ~60% in 2000, 80%+ in 2018
- Why did Taiwan, South Korea keep expanding even when states became less interventionist?

Market dynamics drive capacity

- Customer intimacy (intermediate markets): designs, committed orders and guaranteed capacity between device makers, chip designers and foundries
- Demand responsiveness (end markets): fast product cycles push OEMs, who push chipmakers and fabs to expand

East Asia in the center

- Final assembly, OEMs concentrated there; Asia-Pacific: 71% of 2018 chip demand
- US fabless firms co-locate R&D in E-A, strengthening ties with foundries and competing with Asian companies („being there”)

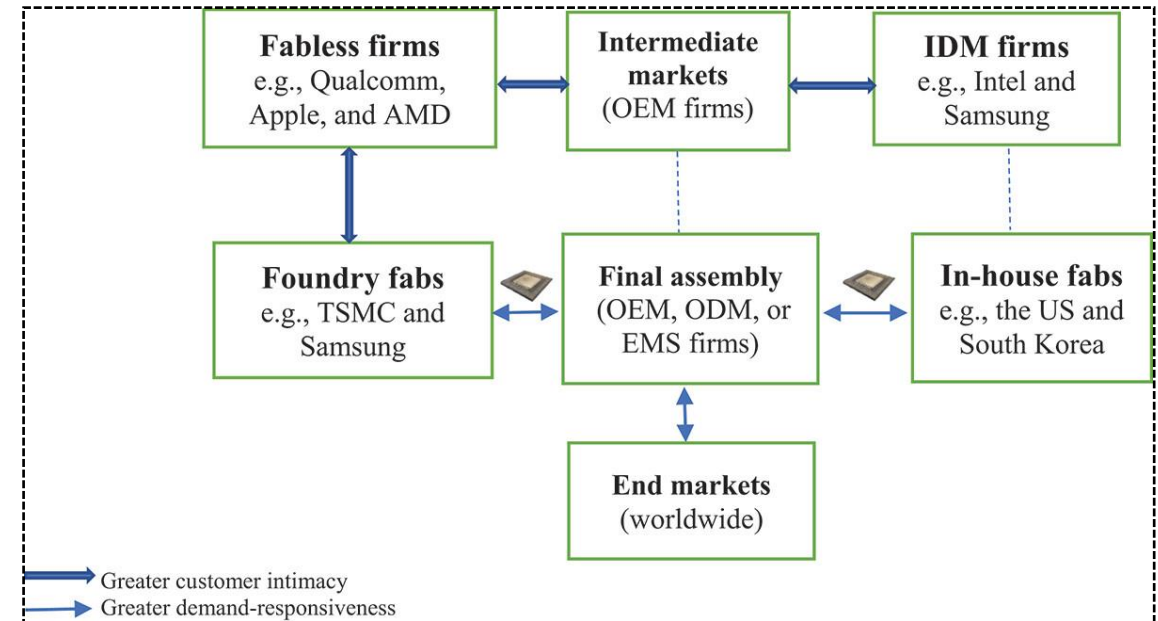


Figure 1: Market dynamics in semiconductor production networks.

Method: fab-level data on 300+ fabs; 20 executive interviews across 15 firms

Conclusion: market dynamics are necessary but not sufficient, they work alongside state support, technology and capital markets

Source: Yeung, H. W. chung. (2022).

Inequalities and polarization

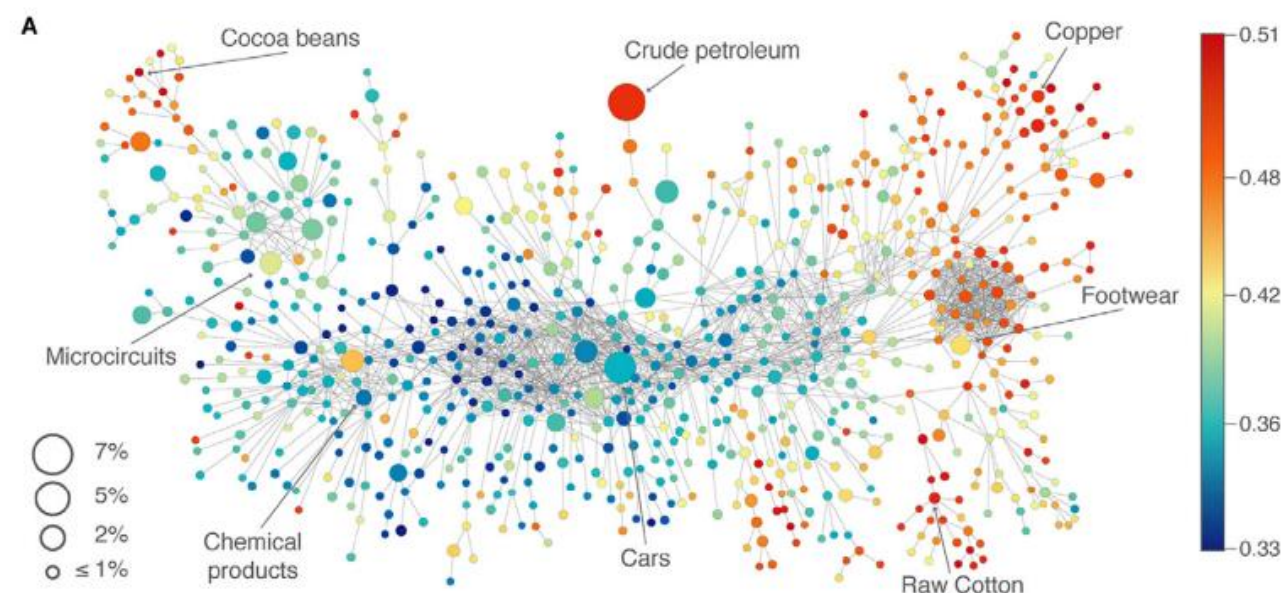
Oct 14

Linking Economic Complexity, Institutions, and Income Inequality (Hartmann et al. 2017)

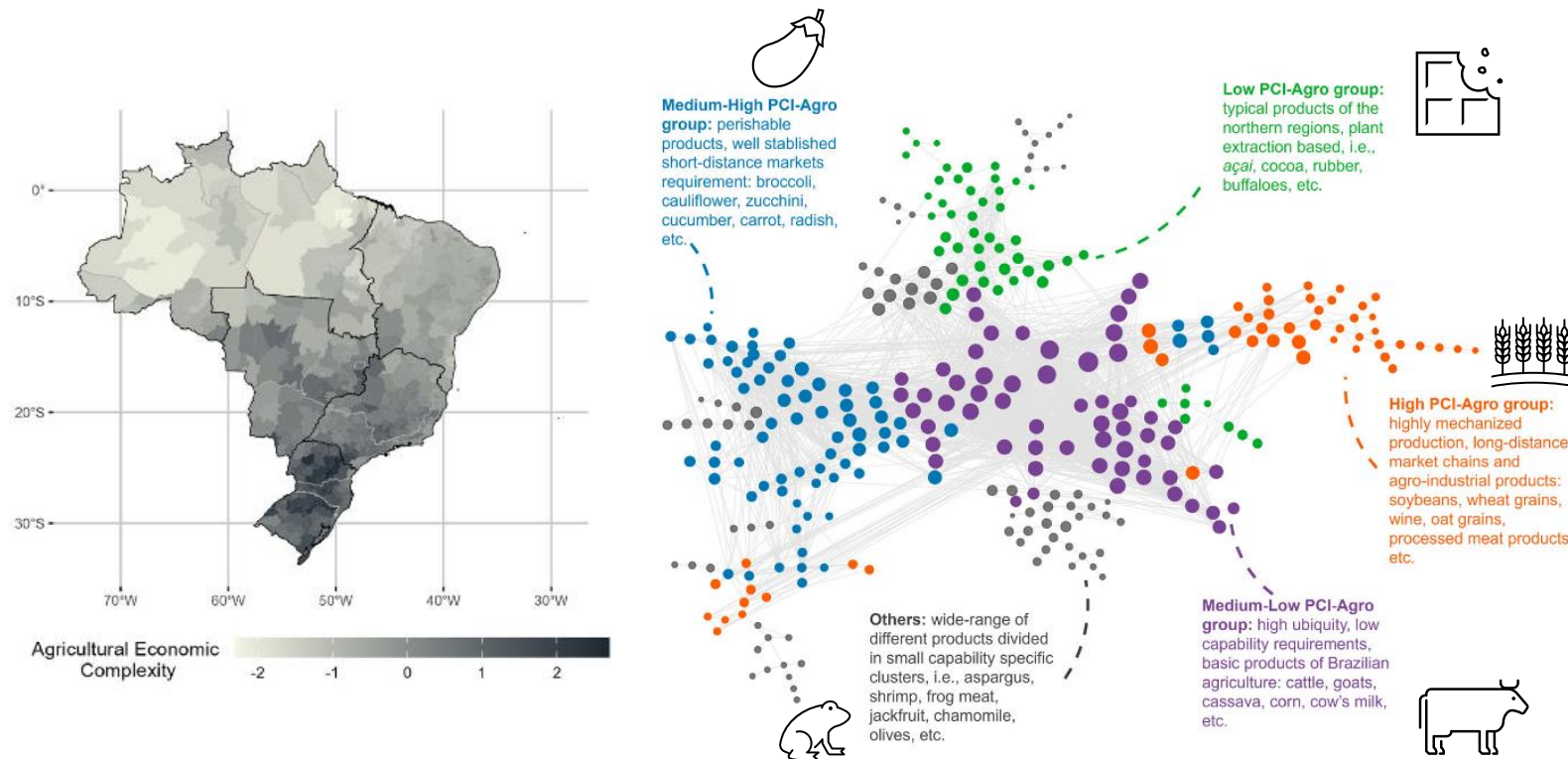
- Topic: Economic growth and inequality.
- Large consensus: the development of productive structures links the two. But how?
- The debate persists due to methodological limitations. A synthesis is attempted by economic complexity as an empirical proxy.
- Hypothesis: Countries exporting high complexity goods (ECI trade Index) on average are less unequal (Gini Index).
- Methodology:
 - Pooled-OLS and FE regression analysis of the hypothesis. Controlled for GDP, schooling and institutions.
 - Product space decomposition for inequality.
- Results:
 - ECI's positive role in economic equality is empirically robust (larger effect size than institutional variable).
 - Product space helps us understand persistent inequality.

Table 3. Fixed-effects panel regression

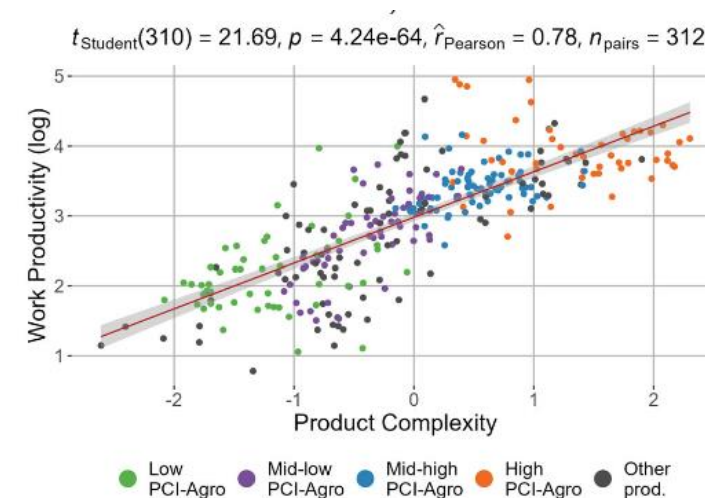
	Dependent variable: GINI						
	I	II	III	IV	V	VI	VII
ECI	-0.031*** -0.007	-0.033*** -0.007	-0.024*** -0.007	-0.026*** -0.007		-0.030*** -0.007	-0.029*** -0.007
ln(GDP PPP pc)		-0.038 -0.028	-0.042 -0.027	-0.017 -0.029	-0.032 -0.03		-0.053* -0.03
ln(GDP PPPpc) ²		0.003* -0.002	0.002 -0.002	-0.00003 -0.002	0.0005 -0.002		0.004** -0.002
Schooling			0.010*** -0.002	0.014*** -0.003	0.015*** -0.003	0.010*** -0.002	
Ln population				-0.024** -0.011	-0.016 -0.011	-0.022** -0.01	0.014* -0.008
Observations	338	338	338	338	338	338	338
R ²	0.077	0.123	0.198	0.213	0.165	0.196	0.134
Adjusted R ²	0.055	0.087	0.139	0.149	0.116	0.138	0.094
F-statistic	20.13*** (df = 1; 240)	11.14*** (df = 3; 238)	14.63*** (df = 4; 237)	12.80*** (df = 5; 236)	11.74*** (df = 4; 237)	19.36*** (df = 3; 238)	9.15*** (df = 4; 237)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes



Agricultural economic complexity and regional inequalities: a new approach using census data from Brazil (Stein et al. 2024)



PCI = Product Complexity Index



DATA: Brazilian Agricultural Census of 2017

More than 5 million rural establishments and 300 products

1. Higher PCI-Agro is associated with higher levels of productivity, education, income and human development
2. ECI-Agro is associated with higher productivity in agriculture
3. 3. Inverted 'U'-shape curve relationship between ECI-Agro and agricultural productivity

THE PERSISTENT EFFECTS OF PERU'S MINING MITA

by Melissa Dell in Econometrica (2010)

Research question

Did Peru's mining *mita* have effects on economic development that persist to this day?

Relevance

identifying how historical institutions can persistently shape economic outcomes

Hypothesis

districts inside the *mita* region are expected to have worse present-day living standards
- effect of less secure land tenure and property rights + weaker provision of public goods

Data

- historical data from the Spanish Empire and Peruvian Republic
- 2001 Peruvian National Household Survey + Ministry of Education height census
- geographic data from NASA SRTM

Methodology

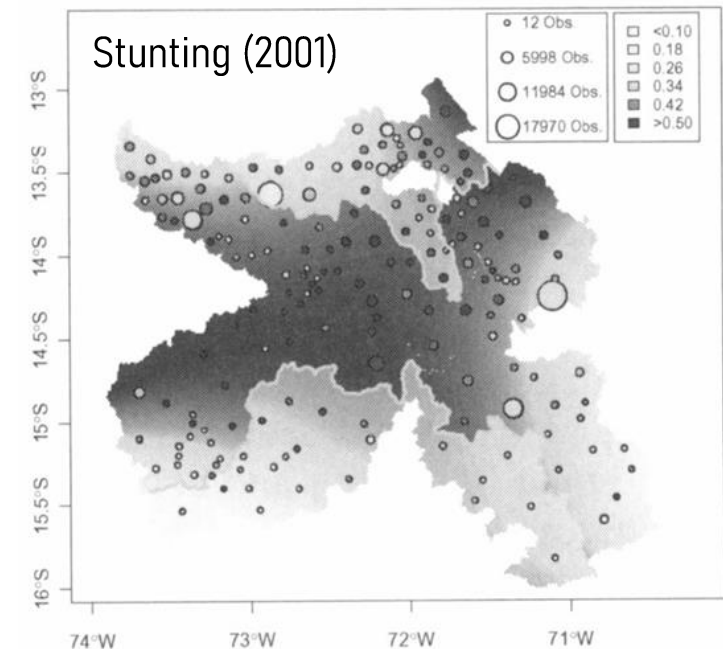
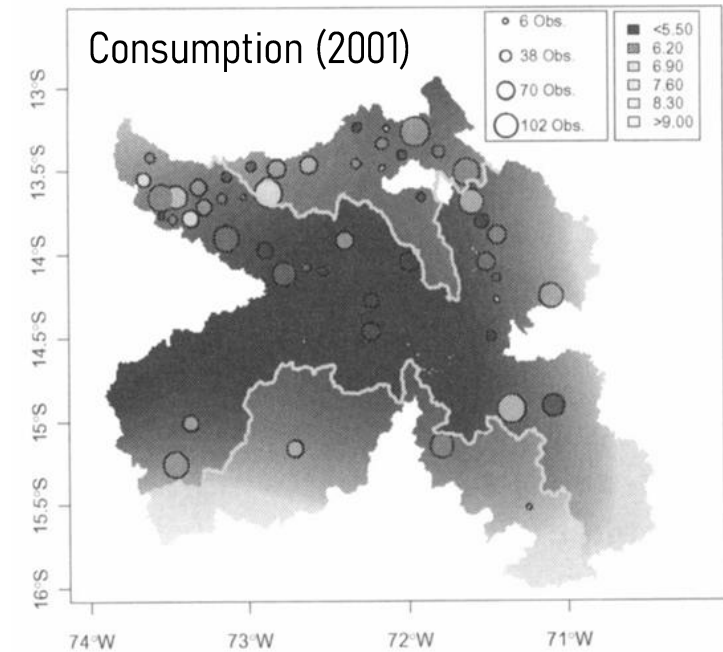
Regression Discontinuity Design (RDD)

comparing districts located just inside and just outside the historical *mita* boundary

Findings

- *mita* districts historically had fewer large landowners and lower educational attainment
 - today are less connected to road networks and more dependent on subsistence farming
- The *mita* reduced household consumption by 25% and increased childhood stunting by 6 pp.

Anna Schenk – Economic Complexity HW1



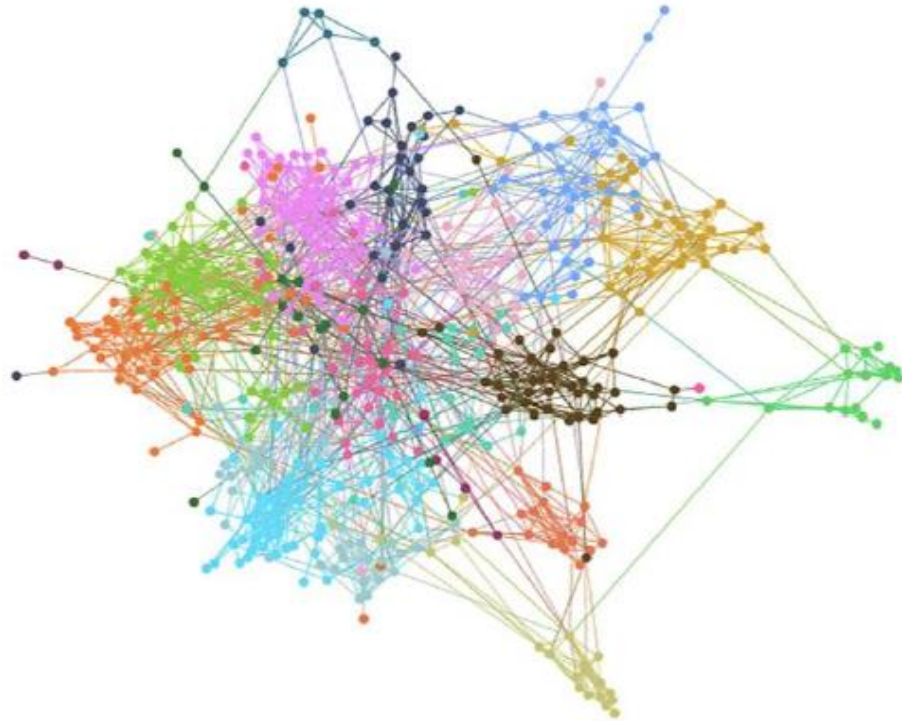
Networks and social capital

Nov 4

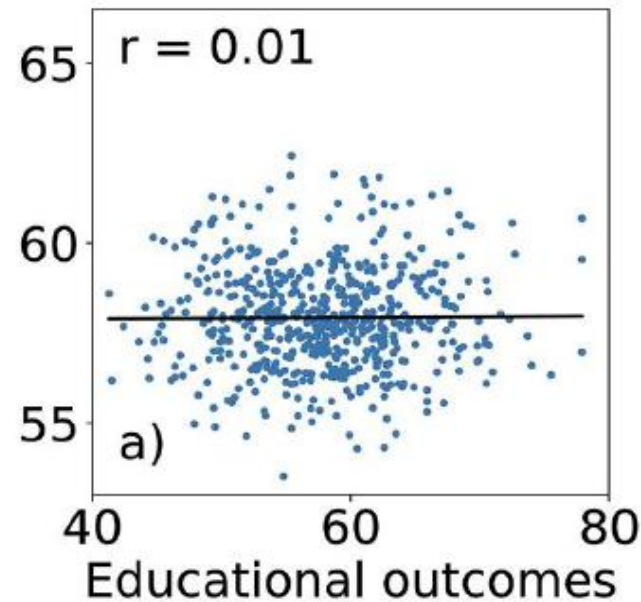
Schools are segregated by educational outcomes in the digital space

Smirnov (2019)

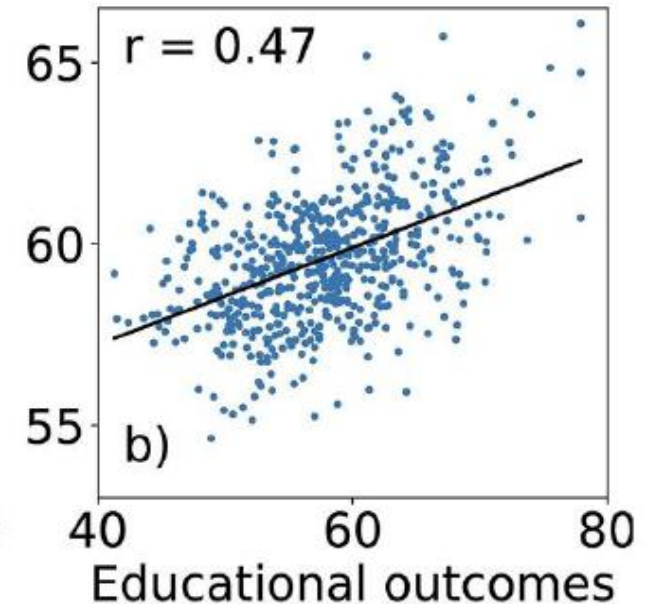
School-friendship network (VK)



Physical neighbours



Digital neighbours



Educational segregation is weak geographically, but strong in the digital space – potentially reinforcing existing inequalities.

Friendships across class lines and upward mobility

Chetty et al. (2022). *Social capital I: Measurement and associations with economic mobility*. *Nature*, 608, 108-121.

PROBLEM

Social capital is thought to shape outcomes from education to health, but it is unclear which forms of social capital are most strongly associated with upward mobility and how they can be measured at scale.

QUESTION

Does social capital (connections between people from different socioeconomic backgrounds) help explain upward economic mobility?

OR

Which forms of social capital are associated with upward income mobility across US areas?

DATA

- 72.2 million Facebook users aged 25-44
- 21 billion friendship connections
- SES (socioeconomic status) predicted by a machine-learning model from proxies
- U.S. ZIP-code and county-level data

METHOD

Three types of social capital, measured by ZIP code and county:

- Cross-type connectedness - economic connectedness (EC): 2 * share of high-SES friends among low-SES people
- Network Cohesiveness - clustering, support ratio, spectral homophily
- Civic engagement - volunteering, civic organizations

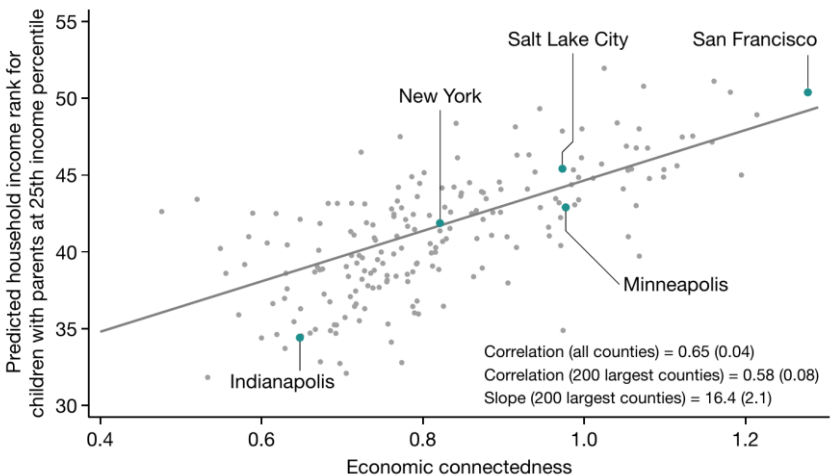


Fig. 4 of the paper: upward mobility vs EC, 200 largest US counties

+20%

If low-SES children had the same level of economic connectedness as high-SES children, their estimated adult incomes would be 20% higher on average.

Correlation with upward mobility

Economic connectedness	0,65
Volunteering rate	0,18
Civic organizations	0,07
Language connectedness	0,06
Penn State index	0,06
Clustering	0,03
Spectral homophily	-0,11
Age connectedness	-0,16
Support ratio	-0,23

County level, Extended Data Table 2

FINDINGS

EC is among the strongest predictors of upward mobility identified to date (correlation = 0.65), while cohesion and civic engagement are not.

Example: Minneapolis (EC 0.98) vs Indianapolis (EC 0.65): low-income children reach the 43rd vs 34th income percentile. EC also accounts for the links of mobility with racial segregation, poverty and inequality.

Course link: bridging ties as channels of opportunity - cf. Tóth et al. (2021) on network segregation and inequality.

Inequality and visibility of wealth in experimental social networks

Akihiro NISHI, Hirokazu SHIRADO, David G. RAND, Nicholas A. CHRISTAKIS (2015, Nature)

THE GAME:



1462 PARTICIPANTS
divided into Erdős-Rényi
random graphs
Average group size: 17.21
Average of 5.33 neighbors
80 sessions, 30 min each



10 ROUNDS, IN EACH ROUND:
COOPERATE (+ 100 units to each of your neighbors, -50 units per neighbor to you);
or DEFECT (no cost, no benefit)
-> population level changes in overall wealth and inequality

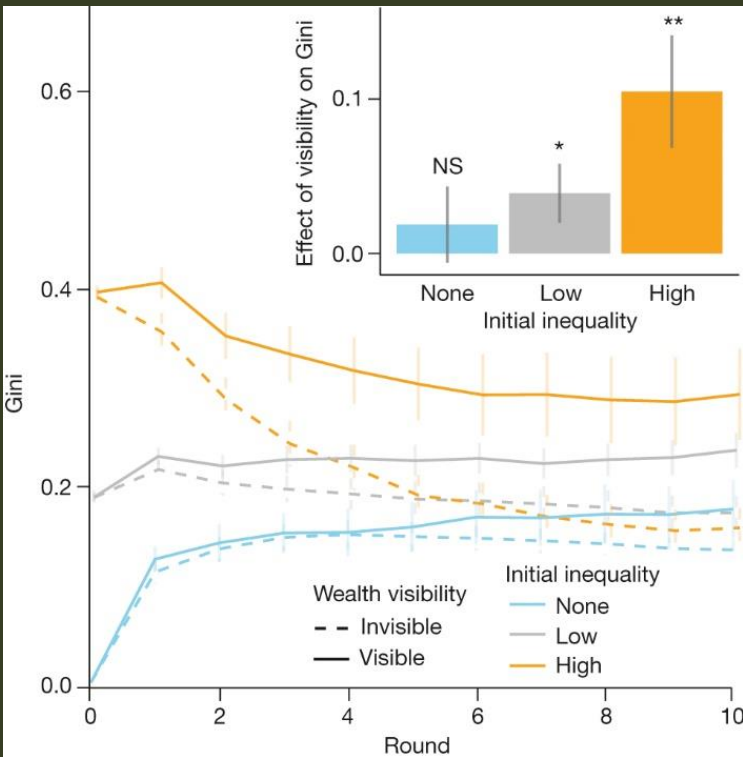
Same average initial wealth; different initial distributions



AFTER MAKING A MOVE, you see your neighbors' moves; randomly chosen 30% of possible connections can be rewired by the affected neighbours (options: connection broken if exists; tied if not; either way left unchanged)

THE RESULTS:

- € wealth **visibility** (not just inequality!) makes pre-existing inequality stick
- € visibility reduced overall **welfare**: lower average wealth, cooperation, and less connected social networks
- € high **initial inequality**: the rich were less likely to cooperate -> the rich gets richer (exploitation scenario)



	Gini-index		
Initial inequality	≈ 0	≈ 0.2	≈ 0.4
Invisible condition*	≈ 0.16	≈ 0.16	≈ 0.16
Visible condition (compared to invisible condition)*	+0.0185	+0.0387	+0.104

*Visibility: whether participants see their neighbours' wealth

Main limitation: simplified lab experiment

Sport and culture

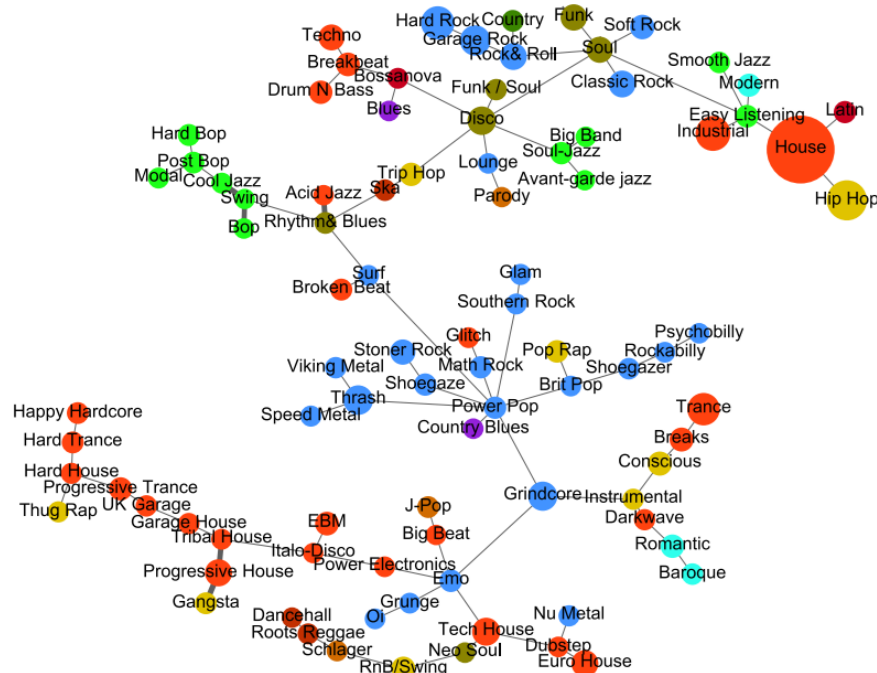
Nov 11

Instrumentational Complexity of Music Genres and Why Simplicity Sells

RESEARCH QUESTION

Can the instruments used in a music style reveal its complexity and how is it related to popularity?

MUSIC STYLE SIMILARITY NETWORK



DATA + METHOD

500K+ albums • 374 styles • 15 genres • 1955-2011

Discogs data connect each style to the instruments and production technologies used on its albums.

Complexity = variety/uniformity

VARIETY

How many instruments?

UNIFORMITY

How specialized are they?

Amazon SalesRank is used for sales.

KEY RESULT

Higher sales -> decreasing complexity

Δ complexity vs. sales: $r = -0.69$, $p = 0.001$

ECONOMIC COMPLEXITY LINK

Instruments are like capabilities, genres are products, and shared instruments create relatedness

TAKEAWAY

Mainstream success is associated with more formulaic instrumentation

Does existing success in related sports increase the probability of winning a country's first medal in a new sport?

DATA

61 768 observations: Summer Olympics (1896-2016) * 61 sport * Competing countries

Binary data: Got a medal: 1 vs Didn't get a medal: 0

METHOD

Relatedness of two sports:

How many countries are "successful,, (have at least 1 medal in) both sports?

Sports Space:

Every node is a sport with related sports as clusters

Relatedness density for a country for a sport:

How many of the related sports to Sport X is Country H already "successful" in?

Entry model:

What makes it more likely that country H wins its first inSport X? (eg. Relatedness density, Competitive balance, Population, GDP, Host, Medals per sport, Portfolio Size etc.)

FINDINGS

Higher relatedness means higher odds of entering a new sport:

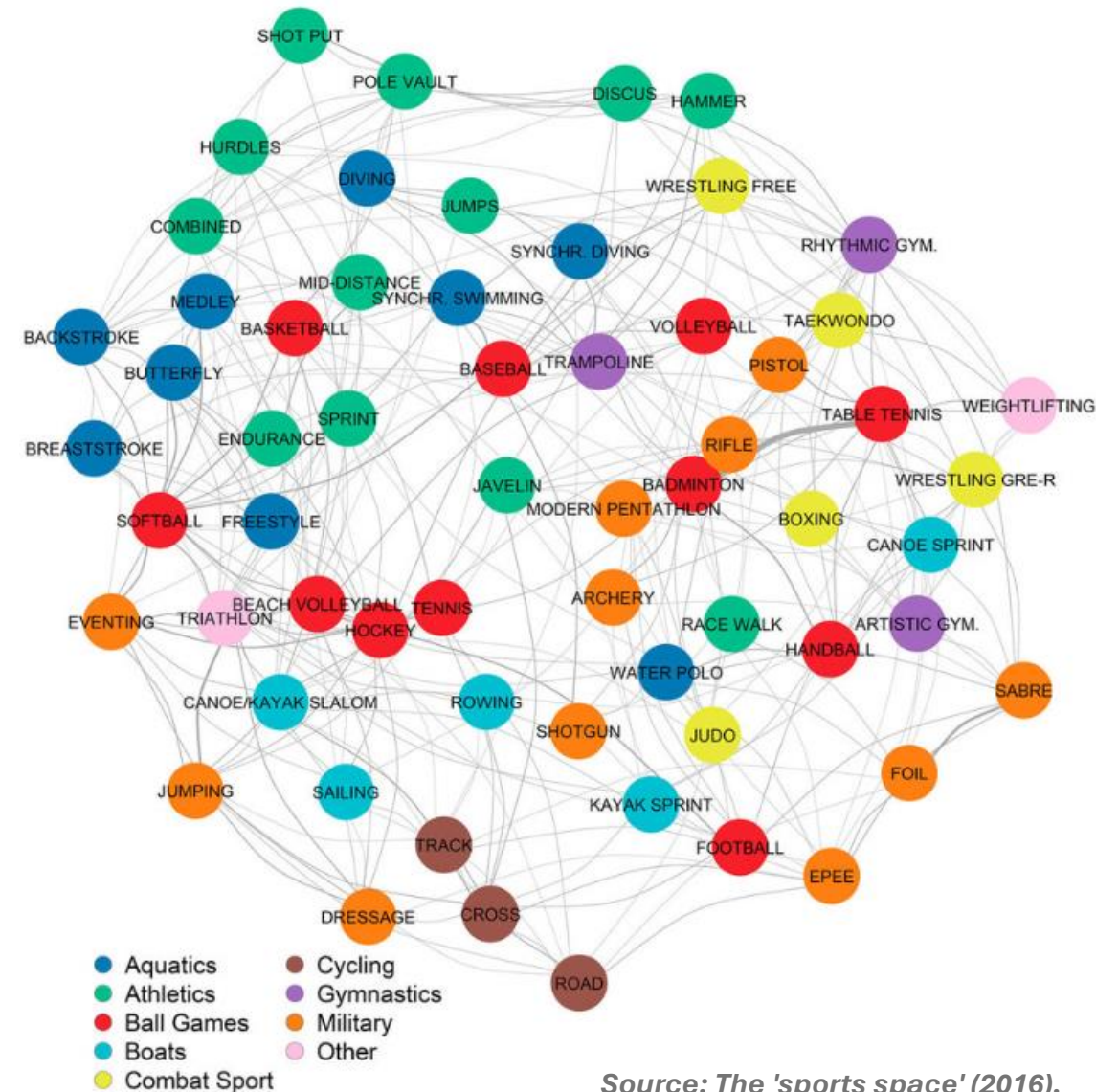
+1 pp relatedness density means ~4% higher odds of entry

„Relatedness is small but robust driving force”:

- Relatively low predictive power
- Outcomes affected by many factors

Source: Does relatedness drive countries' success in new Olympic sports?

Knüpling & Broekel (2022), European Sport Management Quarterly



Source: *The 'sports space' (2016).*
By Tom Broekel

First-mover advantage in music

Sobchuk et al (2024)

?

Why do some songs and musicians become more successful than others? One possible reason could be the „first-mover advantage“. To test this hypothesis, a dataset of over 920,000 songs of 110 genres is analyzed.



Concept

What is **First-mover advantage**?

- Individuals moving to a market niche **early** on („first-movers“) obtain **advantages** compared to those who move to this niche **later**
- If this phenomenon exists in artistic domains **luck** and **quality** can co-exist as explanations for success



Methodology

- Songs Divided into **3 datasets**:
 1. **10 Genres hand-picked** by authors (authors could verify the quality of the genre tags)
 2. **100 Randomly picked genres** to get a broader view
 3. **Control Data Set**: Unstructured in terms of genres, no first-mover advantage expected
- Song's success **standardized** by comparison to all other **songs of the same year** (recency bias)
- Control Data Set divided into „pseudo-genres“, i.e. groups of songs of **various genres**, matching **size** and **yearly distribution** of actual genres
- **Linear models** used to predict playcount based on release year for each dataset



Data Sets

- **Playcounts** from Spotify as margins of success
- **Genre Tags** from Every Noise at Once
- Genres appearing only **before the 2000s excluded (preservation bias)**
- Playcounts **<1000 excluded** (no exact data available)



Results

- The paper finds that **91 out of the 110 genres** show the first-mover advantage.
- For the remaining 19 genres the paper only tries to explain the absence of the effect for one (active suppression of grime music in the 2000s)
- 2 possible mechanism behind first-mover advantage proposed:
 1. **Novelty**: First- movers bring a new style that differentiates them from existing styles
 2. **Canonization**: First-movers often epitomize a genre leading listeners to prioritize their work

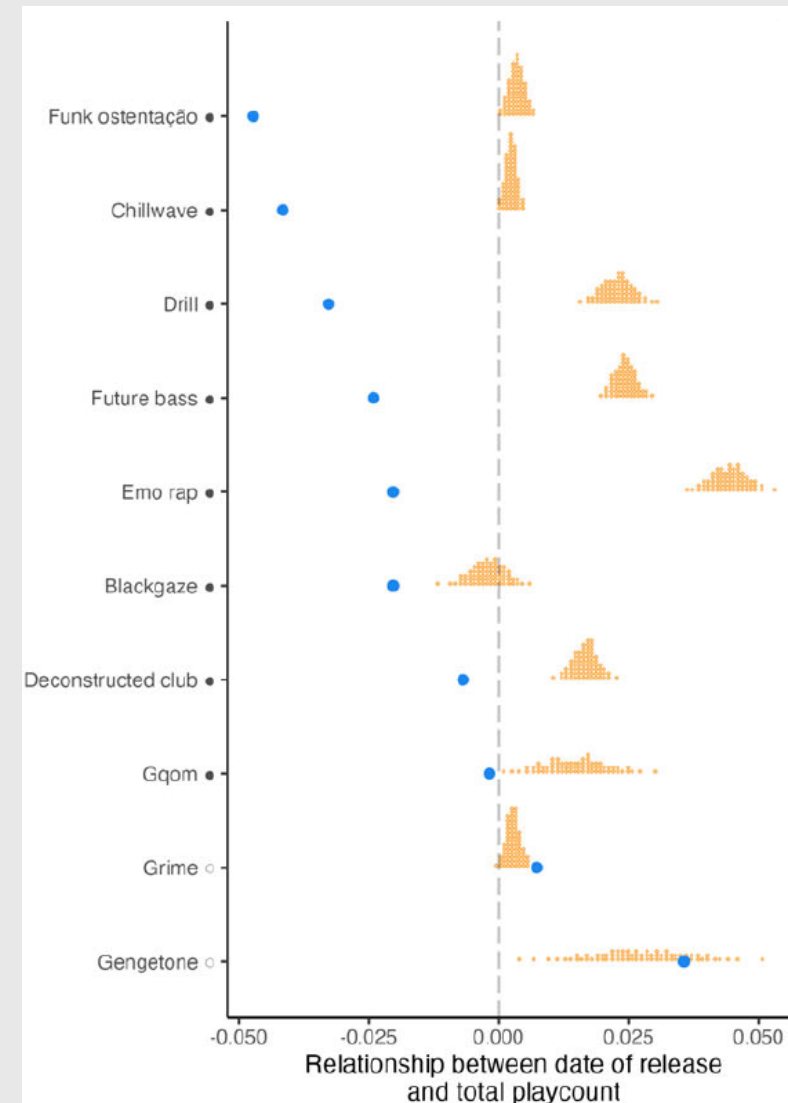
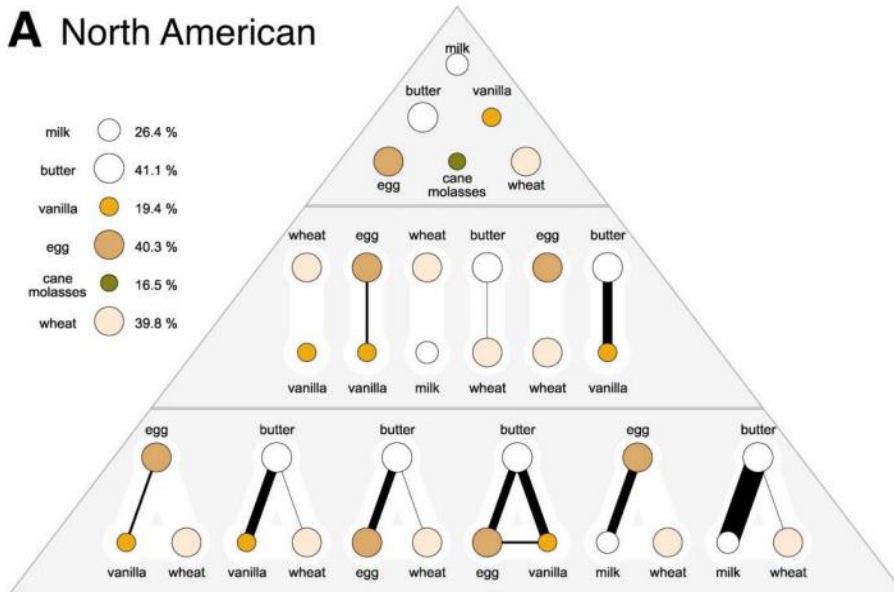


Fig 1: ●: the relationship between time and total playcount for each of the 10 preregistered genres.
: the same relationships, but for the samples drawn from the mixed-genre corpus of songs (1000 samples per each actual genre)
(More negative results indicate first-mover advantage due to transformation)

Food pairing hypothesis across different world cuisines

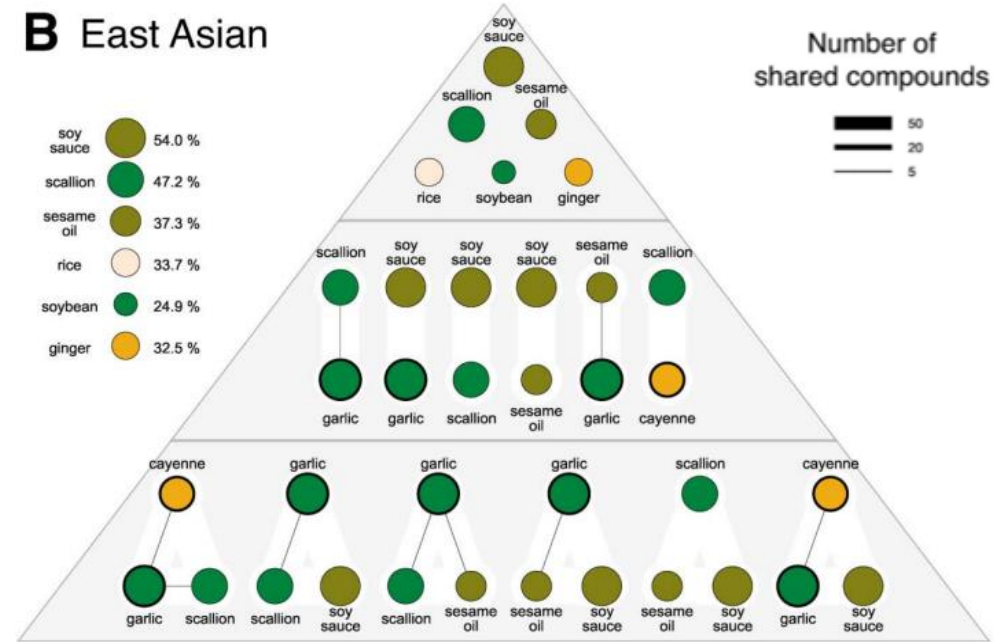
Ahn et al. (2011)

A North American



use recipes whose ingredients
share flavor compounds

B East Asian



avoid recipes whose ingredients
share flavor compounds

no universal rule for pairing flavors

culinary practices are driven by
region and culture

Climate + ecology

Nov 25

Global non-linear effect of temperature on economic production (*Burke, Hsiang & Miguel, Nature 2015*)

Problem: Micro level: Above 20-30 °C, labor productivity and crop yields collapse.

Previous macro view: GDP of rich countries is protected, while that of poor countries decline only linearly.

Research question: Does this protection really exist, or is the macro-effect also nonlinear?

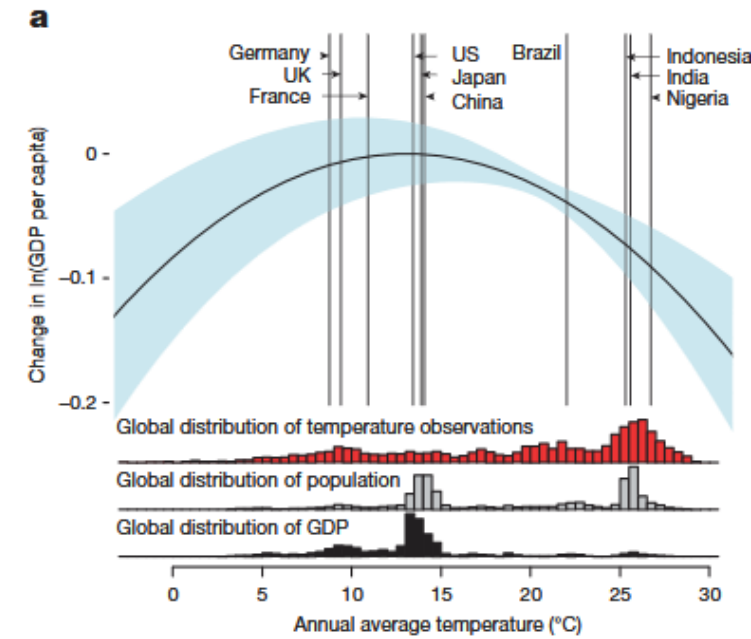
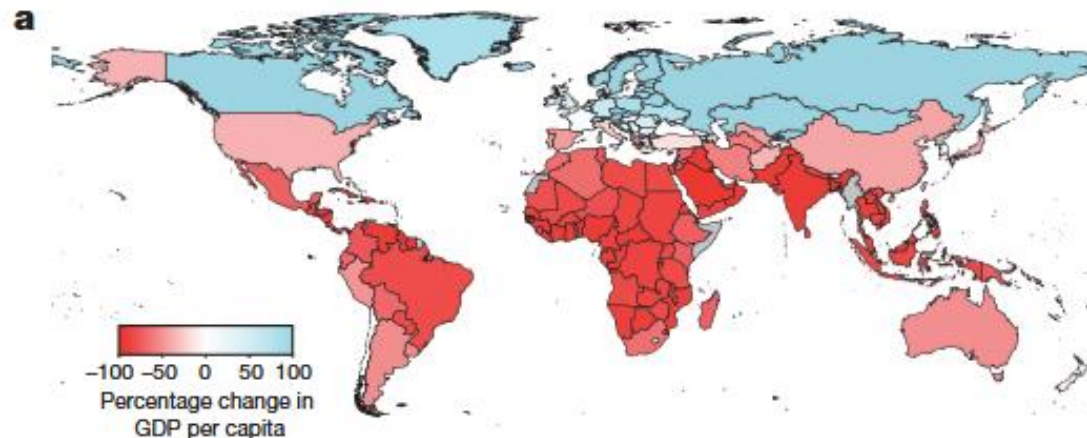
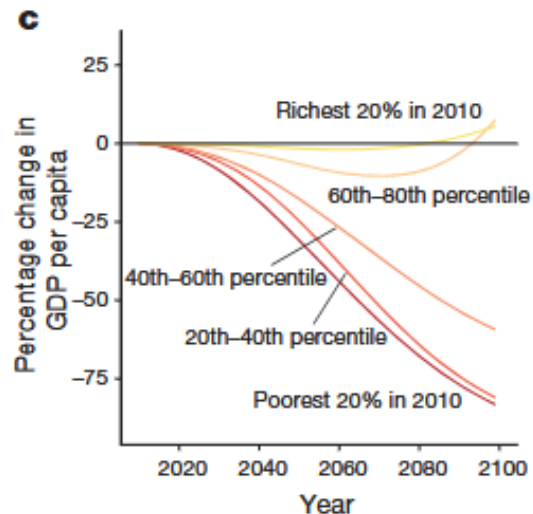
Data: Panel data for 166 countries over the period 1960-2010 (economic data from World Bank, temperature from Udel)

Methodology: Fixed effect regression, non-linear regression

Findings: Non-linear relationship between GDP per capita and average temperature (13 °C is the optimum)

Northern, colder countries may benefit slightly, tropical and southern countries might suffer drastic losses

Inequality could increase between the richest and poorest countries



ECONOMICS FOR THE FUTURE – BEYOND THE SUPERORGANISM

Hagens, N. J. (2020). "Economics for the Future – Beyond the Superorganism." *Ecological Economics*, 169, 106520.

The modern economy should not be studied only through money, GDP, labor, and capital. It also depends fundamentally on: Energy.

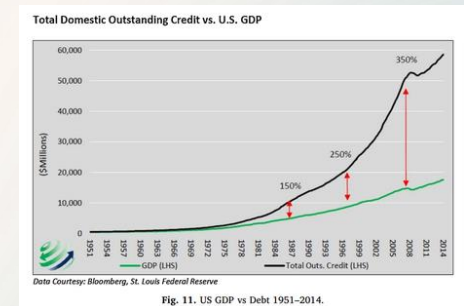
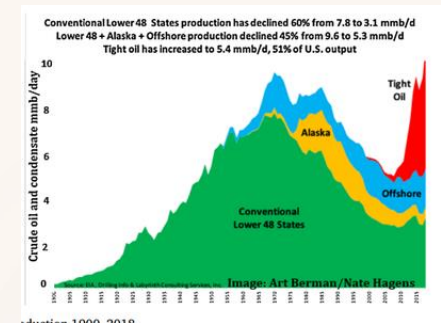
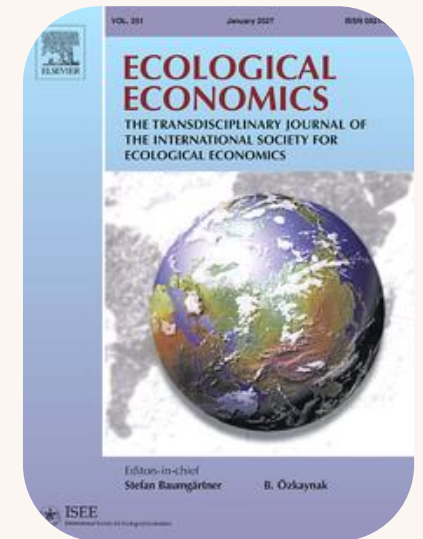


These resources are no longer easily attainable or cheap.

Can economic growth continue indefinitely when it depends on finite energy and material resources?

The paper's argument: Probably not.

Financial systems cannot expand independently of the physical economy: money, production, and debt ultimately depend on energy and material flows.



Why are some countries better at developing green technologies?

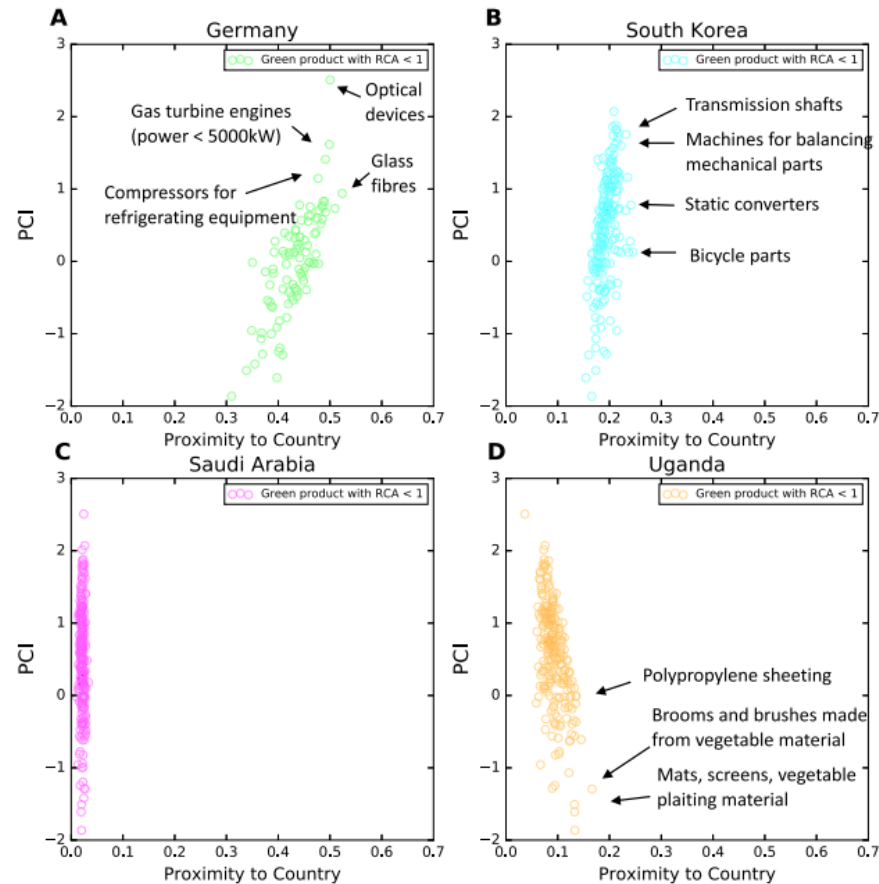


Fig. 6. Illustrating the Green Adjacent Possible for different countries.

In each plot, the circles represent green products that the denoted country is not competitive in. The y-axis plots the PCI of each product and the x-axis plots that product's density to a country. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Mealy & Teytelboym (2020)

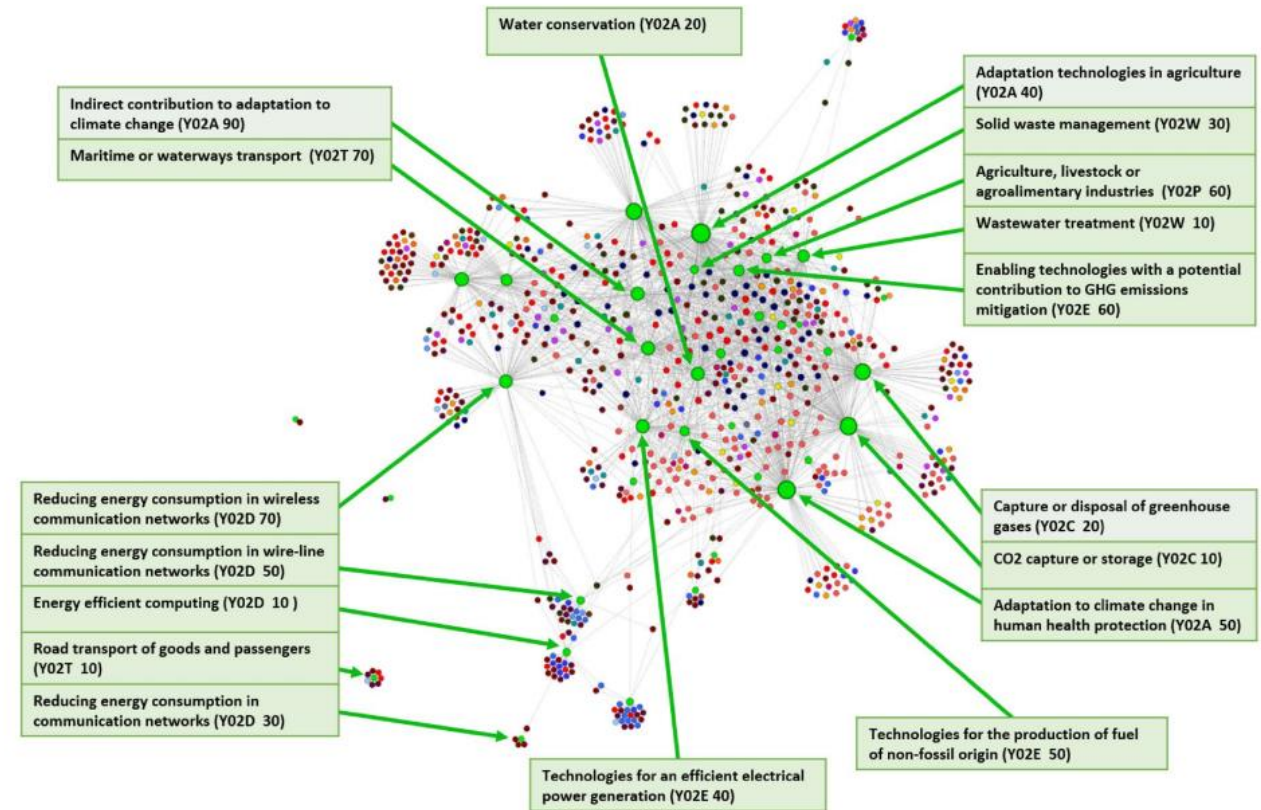


Figure 3. Directed network from green technologies to exported products for a time lag $\Delta T = 0$ and 6-digit products aggregation level. Nodes' size is proportional to their degree. Green nodes: green technologies with green arrows pointing to the description of some of them. All other nodes: exported products (coloured according to Table 2).

de Cunzio et al. (2022)

Zsófia Erdélyi



Vision 2030

لا إله إلا الله محمد رسول الله



Background

- Oil-dependent economy
- Political dependence
- Limited tourism
- Traditional social model

Analysis

- Timeline: 1990-2020
- VECM, dummy variables
- Separated analysis for before and after Vision 2030
- Assymetrical methodology – separate the effects of positive and negative shocks

Results

- Introduction of VAT → role of labour decreasing, role of capita increasing
- Financial development, openness to trade become increasingly important factors
- Impact of oil prices has diminished as a result of tax reforms and restructuring → diversification process

Goals

- Diversifying from oil-dependence
- Developing tourism and logistics
- Increasing competitiveness, becoming a globally integrated economy

Sources: [Article 1](#) & [Article 2](#)